

Answer 4 questions. You should indicate which 4 problems you wish to be graded. Write your solutions clearly in complete English sentences. You may quote theorems (within reason) as long as you state them clearly.

Name: _____

Problems: 1 2 3 4 5 6

1. Let p be a prime and G be a nontrivial finite p -group.
 - (a) (5 points) Prove that $Z(G)$ is not trivial.
 - (b) (5 points) Prove that every nontrivial finite p -group has a normal subgroup of index p .
2. Let $\sigma \in S_n$ be an n -cycle.
 - (a) (2 points) Prove that $\sigma \in A_n$ if and only if n is odd.
 - (b) (4 points) Prove that $C_{S_n}(\sigma) = \langle \sigma \rangle$
 - (c) (4 points) Prove that when n is odd, A_n has two conjugacy classes of n -cycles.
3. Let G be a group and let $T(G) = \{x \in G : \exists n \geq 1, x^n = 1\}$
 - (a) (5 points) Prove that if G is abelian then $T(G)$ is a subgroup of G .
 - (b) (5 points) Give an example of a nonabelian group G such that $T(G)$ is not a subgroup of G . Justify any claims that you make.
4.
 - (a) (5 points) Find the order of the group $\text{GL}(4, \mathbb{Z}/3\mathbb{Z})$.
 - (b) (5 points) Show that there exists a nonabelian group of order $2025 = 3^4 \cdot 5^2$ in which the Sylow 3-subgroup is abelian.
5. Let K and H be groups, with K abelian. Consider a homomorphism $\phi : K \rightarrow \text{Aut}(H)$ and the associated semidirect product $G = H \rtimes_{\phi} K$.
 - (a) (4 points) Prove that when K is identified with the subgroup $\{(1, k) \mid k \in K\}$ of G , $\text{Ker}(\phi)$ is contained in the center of G .
 - (b) (6 points) Prove that every group of order 44 has a cyclic subgroup of order 22.
6. (10 points) Let G be a finite group, and let P be a Sylow p -subgroup of G for some prime p . Let $N = N_G(P)$, the normalizer of P . Prove that $N_G(N) = N$.